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Forward Selection:

Forward selection is a stepwise model-building procedure in regression where variables are added one at a time to the model based on their statistical significance, aiming to maximize the explanatory power while avoiding overfitting.

STEPS

STEP 1: Start with no predictors

- Regress the dependent variable y on each independent variable x_i individually.

$$\hat{y} = \beta_0 + \hat{\beta}_i x_i$$

- Compute SSR for each

$$SSR = \sum (\hat{y} - \bar{y})^2$$

OR

$$\begin{aligned} SSR &= SST - SSE \\ &= \sum (y_i - \bar{y})^2 - \sum (y_i - \hat{y})^2 \end{aligned}$$

- Compute s^2 for each

$$s^2 = \frac{SSE}{n-k-1}$$

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- Select variable with largest SSR
- Evaluate significance of variable with largest SSR using F-test

$$H_0 : \beta_i = 0$$

$$H_1 : \beta_i \neq 0$$

Test stats :

$$F = \frac{R(\beta_i)}{S^2} \rightarrow SSR$$

Critical Region :

$$F_{\alpha}(v_1, v_2)$$

$v_1 = 1$ (because we are adding one variable)

$$v_2 = n - k - 1$$

n : number of observations

k : number of predictors already in the model.

Reject H_0 if

$$F > F_{\alpha}(1, n - k - 1)$$

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STEP 2: Add the next best variable.

Calculate SSR as

$$R(\beta_j | \text{variables already in model}) = \text{SSR}(\text{model with current} + x_i) - \text{SSR}(\text{current model})$$

Choose variable that gives largest SSR

Perform F-test on it

If significant, add it to the model.

FORWARD SELECTION QUESTION:

Table 12.8: Data Relating to Infant Length*

Infant Length, y (cm)	Age, x_1 (days)	Length at Birth, x_2 (cm)	Weight at Birth, x_3 (kg)	Chest Size at Birth, x_4 (cm)
57.5	78	48.2	2.75	29.5
52.8	69	45.5	2.15	26.3
61.3	77	46.3	4.41	32.2
67.0	88	49.0	5.52	36.5
53.5	67	43.0	3.21	27.2
62.7	80	48.0	4.32	27.7
56.2	74	48.0	2.31	28.3
68.5	94	53.0	4.30	30.3
69.2	102	58.0	3.71	28.7

*Data analyzed by the Statistical Consulting Center, Virginia Tech, Blacksburg, Virginia.

STEP 1:

For x_1 ,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$$

$$\hat{y} = 19.0111 + 0.5180 x_1$$

$$R(\beta_1) = SSR(\beta_1) = 288.1468$$

$$s^2 = 4.7276$$

For x_2 ,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_2 x_2$$

$$\hat{y} = 3.4293 + 1.1796 x_2$$

$$R(\beta_2) = 215.3013$$

$$s^2 = 15.1341$$

For x_3 ,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_3 x_3$$

$$= 45.985 + 4.1350 x_3$$

$$R(\beta_3) = 186.1065$$

$$s^2 = 19.3048$$

For x_4 ,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_4 x_4$$

$$= 27.1873 + 1.1399 x_4$$

$$R(\beta_4) = 100.8594$$

$$s^2 = 31.4829$$

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Since $R(\beta_1)$ is largest, select x_1 and calculate its significance.

⇒ Hypothesis:

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

⇒ Test statistics:

$$F = \frac{R(\beta_1)}{S^2}$$

$$= \frac{288.1468}{4.7276}$$

$$= 60.95$$

⇒ Critical Region:

$$\alpha = 0.05$$

$$U_1 = 1$$

$$U_2 = N - K - 1 = 9 - 1 - 1 = 7$$

$$F_{0.05}(1, 7) = 5.59.$$

⇒ Decision:

Test value inside critical region so H_0 Reject

β_1 is significant so keep x_1 .

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STEP 2 = x_1 Fix

For x_1, x_2

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

$$\hat{y} = 44.1004 + 0.9830x_1 - 1.2866x_2$$

$$R(\beta_1, \beta_2) = 312.0172$$

$$\begin{aligned} R(\beta_2 | \beta_1) &= R(\beta_1, \beta_2) - R(\beta_1) \\ &= 312.0172 - 288.14 \\ &= 23.8704 \end{aligned}$$

$$S^2 = 1.537.$$

For x_1, x_3

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3$$

$$\hat{y} = 20.1085 + 0.4136x_1 + 2.0253x_3$$

$$R(\beta_1, \beta_3) = 317.4555$$

$$\begin{aligned} R(\beta_3 | \beta_1) &= 317.45 - 288.14 \\ &= 29.308 \end{aligned}$$

$$S^2 = 0.6307.$$

For x_1, x_4

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_4 x_4$$

$$\hat{y} = 9.3241 + 0.4699x_1 + 0.4582x_4$$

$$R(\beta_1, \beta_4) = 301.9647$$

$$\begin{aligned} R(\beta_4 | \beta_1) &= 301.9647 - 288.14 \\ &= 13.8181 \end{aligned}$$

$$S^2 = 3.2126.$$

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since $R(\beta_3 | \beta_1)$ is largest, select x_3 and evaluate significance.

Hypothesis:

$$H_0 : \beta_3 = 0$$

$$H_1 : \beta_3 \neq 0$$

F-Test:

$$F = \frac{R(\beta_3 | \beta_1)}{S^2}$$

$$= \frac{29.3087}{0.6307}$$

$$= 46.47$$

Critical Region

$$F_{\alpha}(v_1, v_2)$$

$$v_1 = 1$$

$$v_2 = N - K - 1 = 9 - 2 - 1 = 6$$

$$F_{0.05}(1, 6) = 5.99$$

Decision:

Reject H_0 , x_3 is significant. Keep it.

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STEP 3: x_1, x_3 fix

For x_1, x_3, x_2 ,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_2 x_2$$

$$\hat{y} = 5.6298 + 0.0810x_1 + 0.7715x_2 + 8.0694x_3$$

$$R(\beta_1, \beta_2, \beta_3) = 318.2503$$

$$\begin{aligned} R(\beta_2 | \beta_1, \beta_3) &= R(\beta_1, \beta_2, \beta_3) - R(\beta_1, \beta_3) \\ &= 318.250 - 317.45 \\ &= 0.8 \end{aligned}$$

$$S^2 = 0.5979.$$

For x_1, x_3, x_4 ,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4$$

$$\hat{y} = 21.8735 + 0.4128x_1 + 2.2027x_3 - 0.0789x_4$$

$$R(\beta_1, \beta_3, \beta_4) = 317.64$$

$$R(\beta_4 | \beta_1, \beta_3) = 0.19$$

$$S^2 = 0.718.$$

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Since $R(\beta_2 | \beta_1, \beta_3)$ is large,
evaluate significance of x_2

Hypothesis:

$$H_0 : \beta_2 = 0$$

$$H_1 : \beta_2 \neq 0$$

Test Statistics:

$$F = \frac{R(\beta_2 | \beta_1, \beta_3)}{S^2}$$

$$= \frac{0.8}{0.5979}$$

$$= 1.33$$

Critical Region:

$$F_{0.05}(1, 5) = 6.61$$

$$v_1 \quad v_2 = 9 - 3 - 1$$

Decision-

Accept H_0 , β_2 is not
significant, drop x_2 .

Final Model:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3$$

$$\hat{y} = 20.1085 + 0.4136 x_1 + 2.0253 x_3$$